

Bilingual Literature Difficulty Classification Using Multilingual Embeddings

Abstract

This project investigates bilingual literature difficulty classification for English and Chinese literary texts. I construct a bilingual dataset labeled into three readability levels (A/B/C) using a combination of rule-based selection, large language model (LLM)-assisted annotation, and manual calibration. Several multilingual embedding models are evaluated with lightweight classifiers to determine their effectiveness for cross-lingual readability prediction. Experimental results show that multilingual sentence embeddings can achieve strong classification performance across both languages, while modern transformer-based embeddings outperform traditional baselines. The project also includes a HuggingFace web demo for interactive prediction.

1. Introduction

Automatic readability assessment is an important task in Natural Language Processing (NLP), with applications in education, language learning, and digital humanities. While readability classification has been extensively studied for English, bilingual literary readability classification remains relatively underexplored. This project uses multilingual embeddings to build a bilingual literature difficulty classifier that categorizes literary excerpts into three levels aligned with the Common European Framework of Reference (CEFR). The main contribution is the construction of a bilingual literature difficulty dataset using LLM-assisted annotation and multilingual embedding-based classification.

2. Dataset Construction

The dataset consists of English and Chinese literary excerpts collected from public-domain books and online literature sources. English texts include 39 classical and modern literature, while Chinese texts contain 9 modern literary works.

Texts were segmented into 100-200 token units to maintain contextual meaning. We assigned difficulty levels (A/B/C) based on a comprehensive labeling framework including **lexical**, **syntactic**, and **stylistic features**. The dataset underwent manual review to rectify noisy labels and was supplemented with targeted samples to address class imbalance in lower-volume categories. The final dataset was divided into balanced training and testing sets for both languages.

3. Methodology

The classification pipeline contains three stages: 1) Text preprocessing; 2) Embedding generation comparison; 3) Difficulty classification using lightweight machine learning models.

After creating the standard train and test datasets, I evaluated three multilingual embedding models, including paraphrase-multilingual-MiniLM-L12-v2, and xlm-roberta-base and LaBSE embeddings. The comparison of three embedding models revealed paraphrase-multilingual-MiniLM performed the best, and thus was chosen for the comparison of three classifiers, including logistic regression, SVM linear and transformer mlp.

Both the Performances of embedding and classifiers are evaluated using Accuracy and Macro F1-score.

Macro F1 is particularly important because it balances performance across all difficulty classes.

4. Experimental Results

Table below summarizes the performance of several multilingual embedding models on the bilingual test set.

Model	Accuracy	Macro F1
TF-IDF + Logistic Regression	0.5552	0.5409
TF-IDF +SVM	0.5798	0.5712
paraphrase-multilingual-MiniLM + Transformer	0.5767	0.5751

The results show that the traditional SVM model achieved the highest score in accuracy and transformer-based multilingual embeddings slightly outperformed traditional sparse-text baselines. Error analysis suggests that medium-difficulty (B-level) excerpts are the most difficult to classify due to overlap between simplified literary language and moderately complex narrative structures.

5. Discussion

The experiments demonstrate that multilingual embeddings can generalize effectively across English and Chinese literary texts despite major linguistic differences. Semantic embedding models appear particularly robust for handling diverse literature and stylistic variation.

However, several limitations remain. First, some labels generated through LLM-assisted annotation may still deviate from the true literature difficulties perceived by human readers as what the embeddings capture cannot equal to true reading experience. Second, the dataset size is relatively limited compared with large-scale readability corpora, which partially accounts for the competitive result by SVM. Future work may explore larger datasets, hierarchical difficulty scoring, and cross-lingual transfer learning.

6. Conclusion

This project presents a bilingual literature difficulty classification system using multilingual embeddings and LLM-assisted dataset construction. Experimental results indicate that modern multilingual transformer embeddings provide strong performance for cross-lingual readability classification tasks. The project demonstrates the feasibility of combining multilingual NLP techniques with lightweight classifiers for educational and literary analysis applications.

7. Reflection on AI-Assisted Development

It is undeniable that vibe coding can significantly boost the efficiency of development of NLP project, especially for rapid prototyping, dataset construction, and code generation. However, it also required substantial computational resources and strong project management skills. During the early stages of my project, multiple iterations were needed for dataset design, text collection, and difficulty annotation, combining AI assistance with my background in linguistics and English teaching. Also, generating and refining multiple versions of the code consumed a large number of tokens and computational resources, which later became a limitation. In practice, AI-assisted development still requires a solid understanding of machine learning pipelines, models, and evaluation methods; otherwise, considerable time may still be spent on debugging, retraining, and model selection. Overall, AI tools provide an accessible and highly productive workflow, particularly for users without a formal computer science

background.

8. Project Resources/Links

- GitHub Repository: <https://github.com/merriam-tao/BiLit-CEFR>
- HuggingFace Dataset: <https://huggingface.co/datasets/merriamtao/BiLit-CEFR>
- HuggingFace Demo: <https://huggingface.co/spaces/merriamtao/bilit-cefr-demo>